

Lewis Structures (Kelter) — step 0 for (1–8)

1. Sum valence e (= group #; add 1 per -, remove 1 per + for ions)
2. Sum "wants" (table; period 3+ central may exceed 8)
3. Bonds = (wants - valence) ÷ 2 (round down)
4. Central atom = lowest EN (**never H**); ions go in brackets, charge outside
5. Remaining e = valence - 2×bonds: outer atoms first, then central; double/triple bond if central lacks octet or nonzero FC
6. FC = valence - (lone e + # bonds): want overall 0, else - on most EN atom

	H	Be	B, Al	C, Si	N, P	O, S	F-I	Noble
Val e	1	2	3	4	5	6	7	8
Wants	2	4	6	8	8	8	8	8

Periodic Trends

	Trend	Top	Bot	Left	Right
Atomic size (radius)	↘ (Cs)	↓	↑	↑	↓
EN (electronegativity)	↗ (F)	↑	↓	↓	↑
IE (ionization energy)	↗ (He)	↑	↓	↓	↑
EA (electron affinity)	↗ (Cl)	↑	↓	↓	↑

- Cation < neutral < anion; $\uparrow \Delta \text{EN} \rightarrow \uparrow$ bond dipole (3–4); smaller ion, higher charge $\rightarrow \uparrow \Delta H_{\text{hydr}}$ (18); bigger $\rightarrow \uparrow$ dispersion (9–12)

(1–2) Molecular Geometry (VSEPR)

- Central atom BG + LP (double/triple bond or radical e = 1 group)

e ⁻	Hyb	BG	LP	Mol. geo	e ⁻ geo	Angle	Pol
2	sp	2	0	linear	linear	180°	X
3	sp ²	3	0	tri. planar	tri. planar	120°	X
3	sp ²	2	1	bent	tri. planar	<120°	✓
4	sp ³	4	0	tetra.	tetra.	109.5°	X
4	sp ³	3	1	tri. pyr.	tetra.	<109.5°	✓
4	sp ³	2	2	bent	tetra.	<109.5°	✓
5	sp ³ d	5	0	tri. bipy.	tri. bipy.	90/120°	X
5	sp ³ d	4	1	seesaw	tri. bipy.	90/120°	✓
5	sp ³ d	3	2	T-shaped	tri. bipy.	90, 180°	✓
5	sp ³ d	2	3	linear	tri. bipy.	180°	X
6	sp ³ d ²	6	0	octa.	octa.	90°	X
6	sp ³ d ²	5	1	sq. pyr.	octa.	90°	✓
6	sp ³ d ²	4	2	sq. planar	octa.	90°	X

- Lone pairs squeeze angles (SCl₂ a little < 109.5°). SO₂, BF₃ tri. planar; PCl₅, PH₃, NH₃ tri. pyr.; CS₂, CO₂, N₂, I₂ linear; SiH₄, CH₄, PO₄³⁻ tetra.; SCl₂, H₂S, H₂O, O₃, ClNO, NH₃ bent; ClF₃ T-shaped; XeF₄ sq. planar

(3–4) Polarity & Dipole Moment

- **Dipole** = separation of + / - charge; net dipole moment ($\mu \neq 0$) = polar molecule
- Polar = polar bonds (ΔEN) whose vectors **don't cancel**. Pol column above assumes identical outer atoms — **mixed outer atoms** → polar (CH₃Cl, CHCl₃, CHCl₂)
- Nonpolar: same-element diatomic (Br₂), all hydrocarbons, X-row shapes: XeF₄, XeF₂, CCl₄, PBr₃, PCl₃, BeCl₂, CO, BF₃, BH₃, SO₂
- Polar: different-element diatomic (CO, IBr, HCl) + lone-pair shapes: BrF₃ (sq. pyr.), NH₃, OF₂, H₂O, SO₂ (bent), NCl₃, NF₃
- $\uparrow \Delta \text{EN} = \uparrow \mu$: CH₃F > CH₃Cl > CH₃Br > CH₃I; NH₃ > PH₃ > AsH₃ (EN falls down group)
- Polar dissolves polar: NH₃ in H₂O ✓, H₂S in NH₃ ✓, CCl₄/SiO₂/O₂/CO₂ in polar X

(5–6) Hybridization of an Atom in a Structure

- Count e⁻ groups **on that atom** (bonded atoms + lone pairs) → table (1): 2 sp, 3 sp², 4 sp³, 5 sp³d, 6 sp³d²
- sp² geometry is trigonal planar (**not** square planar)
- C, 4 single bonds → **sp³**; C with 1 double bond (C=O, C=C, benzene ring) → **sp²**; C with triple or 2 doubles → **sp**
- O in -O-H or ether (2 atoms + 2 LP) → **sp³**; carbonyl =O → **sp²**
- N with 3 atoms + 1 LP (amine) → **sp³**
- Each atom separately: CH₃-CHO → left C sp³, carbonyl C sp²

(7–8) σ and π Bonds: Counting & Concepts

- **single = 1 σ ; double = 1 σ + 1 π ; triple = 1 σ + 2 π**
- Count **every** bond, incl. each C-H! σ = # of atom-atom connections; π = extras on doubles/triples
- CH₃-CH=CH-COOH: 11 σ , 2 π ; benzene C₆H₆: 12 σ (6 C-C + 6 C-H) + 3 π
- σ : end-to-end overlap, e⁻ density **on** internuclear axis, rotation free, stronger
- π : **side-by-side p-p** overlap, density above & below axis → **restricted rotation** (energy needed to break the p-p interaction) → cis/trans in C=C (1,2-dichloroethene), none around C-C single (1,2-dichloroethane)
- Two 1s (or head-on p+p along bond axis) → σ and σ^* ; sideways p+p (perpendicular axis) → π and π^*
- Benzene delocalized π MOs: e⁻ **free to move around the ring**; all C-C equal

(9–10) Intermolecular Forces

Force	Who has it
Dispersion (London, van der Waals, induced dipole)	ALL molecules; $\uparrow$ w/ size & molar mass; straight-chain > branched
Dipole-dipole	polar molecules
H-bond	H bonded directly to N, O, F
Ion-dipole	ion + polar solvent (NaCl in H ₂ O)

- Dispersion only (nonpolar): Br₂, Cl₂, F₂, H₂, CO₂, BCl₃, CF₄, hydrocarbons, noble gases
- Disp + dip-dip: HCl, HBr, SO₂, CHCl₃, CH₃Br, ethers (CH₃OCH₃), CH₃CHO, CH₃CN
- + H-bond: H₂O, NH₃, HF, CH₃OH, CH₃CH₂OH, CH₃NH₂, CH₃COOH, H₂O₂, HOCH₂CH₂OH
- Traps: H-bond **with water** just needs N/O/F (CH₃COOH ✓, CH₃NH₂ ✓; C₂H₆ X; Na⁺ = ion-dipole); S never H-bonds (H₂S, CH₃SCH₃); ether O has no H → none with itself; polar **bonds** ≠ polar molecule (CO₂)

(11–12) Rank Vapor Pressure & Boiling Point

- Strength: **ionic > H-bond > dipole-dipole > dispersion**, but size can trump: big nonpolar can beat small polar
- $\uparrow \text{IMF} \rightarrow \uparrow \text{bp}, \uparrow \text{mp}, \uparrow \text{viscosity}, \uparrow \text{surf. tension}, \uparrow \Delta H_{\text{vap}}, \uparrow \text{critical } T, \downarrow \text{vapor pressure}$
- **VP order = reverse of bp order** (lowest bp = highest VP at room T)
- Weak IMF ⇒ low ΔH_{vap} , high VP, low bp

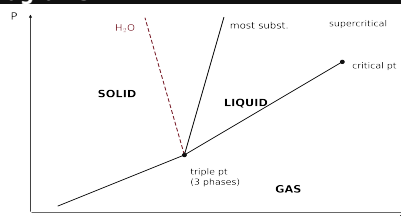
VP ↓	CF ₄ > PF ₅ > BrF ₃ (polar last)
bp ↑	CH ₄ < CH ₃ Cl < CH ₃ OH < RbCl (ionic)
bp (disp. size)	I ₂ > Br ₂ > Cl ₂ > F ₂ ; CH ₃ CHO > CH ₃ OCH ₃ > Rn
bp (H-bond)	HF > HI > HBr > HCl
bp (# of OH)	HOCH ₂ CH ₂ OH > ClCH ₂ CH ₂ OH > CH ₃ CH ₂ OH > CH ₃ CH ₂ Cl

(13) Viscosity, Capillary Action, Surface Tension

Surface tension	energy required to stretch/increase surface area by unit amount
Viscosity	resistance of a liquid to flow ; $\uparrow$ with IMF, $\downarrow$ with $\uparrow T$; spherical/compact → lower
Capillary action	liquid flows up a thin tube against gravity = cohesion (like molecules) + adhesion (liquid-glass)

- Concave meniscus (H₂O): adhesion > cohesion; convex (Hg): cohesion > adhesion. Highest viscosity = most H-bonding (HOCH₂CH₂OH)

(14) Phase Diagrams



- Lines = 2 phases coexist; **triple point** = all 3; **critical point**: above T_c no pressure can liquefy gas → **supercritical fluid**
- $\uparrow T$ at const P (→): solid → liquid → gas. $\uparrow P$ at const T ($\uparrow$): gas → liquid → solid. Normal bp/mp read at 1 atm
- **H₂O**: fusion line leans **left** (ice less dense) → $\uparrow P$ melts ice. **CO₂**: triple pt at 5.1 atm → at 1 atm solid **sublimes** s→g (dry ice)

(15–16) Type of Solid

Type	Held by	Properties	Examples
Ionic	± attraction	hard, brittle, high mp, poor cond.	NaCl, AgCl, CaF ₂ , MgO, CaCO ₃
Molecular	IMFs	soft, low mp, poor cond.	ice H ₂ O, CO ₂ , I ₂ , sucrose
Metallic (atomic)	e ⁻ sea	low-high mp, good cond.	Au, Cu, Fe, Na
Network covalent (atomic)	covalent bonds	v. hard, v. high mp, poor cond.	diamond, quartz SiO ₂ , graphite, SiC
Nonbonding (atomic)	dispersion	v. low mp	noble gases: Ar(s)
Amorphous	no long-range order	melts over a range	glass, rubber, plastic, butter, wax
Ceramic	inorganic nonmetal; powder + H ₂ O + heat	brittle, heat-resistant	clay/kaolinite, porcelain, Al ₂ O ₃

- Graphite sp² sheets: conducts along sheets, slippery; diamond sp³: insulator; fcc packing = most space-efficient (12 neighbors)

(17) Band Theory & Doping

- Bands = MOs delocalized over the whole crystal: **valence band = bonding MOs, conduction band = antibonding MOs, band gap** between
- Conductor** (metal): bands continuous/overlap; **Semiconductor** (metalloids Si, Ge): small gap, e⁻ promotable; **Insulator: LARGE gap**
- Doping** = add impurity to ↑ conductivity:

	Dopant	Carrier
n-type	Group V/5A (P, As)	extra e ⁻ in conduction band (– charges)
p-type	Group III/3A (B, Al, Ga, In)	holes in valence band (e ⁻ hop between holes)
p-n junction	—	diode : current flows one way

(18) Heat of Solution & Hydration

$$\Delta H_{\text{solution}} = \Delta H_{\text{hydration}} - \Delta H_{\text{lattice energy}}$$

- ΔHsolution**: heat absorbed/released on dissolving = ΔH(solvent-solvent) + ΔH(solute-solute) + ΔH(solvent-solute)
- ΔHhydration**: heat **released** when 1 mol **gaseous ions** dissolves in water (forms ion-dipole; always exo). ΔHsolute = –ΔHlattice
- Highest ΔHhydr**: smallest ion + highest charge: Al³⁺ > Mg²⁺ > Na⁺ > K⁺; F > Cl⁻
- ΔHsoln **exo (–)**: ΔHhydration > ΔHsolute; **endo (+)**: solute > hydration; **≈ 0**: comparable

(19) Saturation; Like Dissolves Like

Unsaturated	less than equilibrium amount — more can dissolve
Saturated	max amount at that T; dissolved ⇌ solid dynamic equilibrium
Supersaturated	more than equilibrium amount; unstable — seed crystal/scratch → excess crystallizes out

- Solution forms if solute-solvent attraction **comparable** to solute-solute + solvent-solvent; polar/ionic ↔ polar (H₂O, ethanol, NH₃); nonpolar ↔ nonpolar (hexane, CCl₄, benzene)
- Most soluble in ethanol = most H-bonds (ethylene glycol); in hexane = longest chain (1-pentanol); all **nitrates** soluble

(20) Solubility vs P & T; When Bubbles/Boiling

↑ gives	Gas	Solid / liquid
↑ Pressure	↑ solubility (Henry)	no effect
↑ Temperature	↓ solubility	↑ solubility

- Dissolve the most gas: **low T + high P** (cold soda holds CO₂)
- Liquid **boils** when its vapor pressure = **external (atm) pressure**; gas **bubbles out** of solution when its partial (vapor) pressure > atm pressure

(21) Molarity (and Molality)

$$M = \frac{\text{mol solute}}{\text{L solution}} \quad m = \frac{\text{mol solute}}{\text{kg solvent}}$$

- mol = M × V(L); V(L) = mol ÷ M; mL ÷ 1000 = L; g ↔ mol via molar mass; dilution: M₁V₁ = M₂V₂
- 0.250 L of 0.500 M₁ → 0.125 mol; need 0.300 mol of 0.450 M → 0.667 L = 667 mL
- Molality m uses **kg of solvent**, not solution — for ΔTf/ΔTb. Only molarity changes with T (volume swells); m, mass %, χ don't

(22) Percent Concentration (% w/v)

$$\% w/v = \frac{g \text{ solute}}{100 \text{ mL soln}} \Rightarrow g = \frac{\% \times mL}{100}$$

- 1.5% (w/v), 250 mL → 1.5 × 250/100 = **3.75 g** (n.b. the summary's "31.25 g" contradicts its own formula — trust g = % × mL / 100)
- mL from % w/v: 16.6 g ÷ 0.0450 g/mL = 369 mL
- % m/m** = g solute / (g solute + g solvent) × 100; g water = g soln × (1 – %/100): 300 g of 9.71% KBr → 271 g water; **% v/v** = vol / total vol × 100

(23) Henry's Law (Cgas = kH·Pgases)

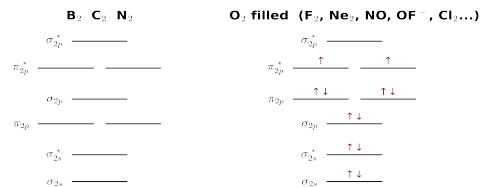
$$C_{\text{gas}} = k_H P_{\text{gas}} \quad k_H = C_{\text{gas}} / P_{\text{gas}}$$

- kH = Henry's constant = "solubility constant of a gas" (M/atm); C = molar solubility
- mmHg → atm: ÷ **760**
- kH = C/P: 1.3 M at 0.022 atm → **59 M/atm**
- 4.7e–4 M at 522 mmHg → kH = 4.7e–4 ÷ (522/760) = **6.8e–4 M/atm**

(24) Osmosis

- Osmosis**: flow of **solvent** (water) across a semipermeable membrane from **low solute conc.** → **high solute conc.** (dilute → concentrated)
- Semipermeable membrane**: passes solvent, **blocks solute** (ions) — it does *not* stop water
- Hyperosmotic** = more solute: water flows **into** it (cell in it shrinks/crenates)
- Hyposmotic** = less solute: water flows **out** into the stronger side (cell in it swells/bursts)
- Isosmotic** = equal: no net flow (isotonic saline = 0.9% w/v NaCl); n = pressure needed to **stop** osmotic flow

(25) WORD: MO Theory — Bond Order & Magnetism

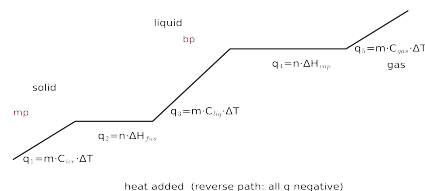


- Valence e⁻ total: **add 1 per – charge, remove 1 per +**
- Fill low → high, 2 per orbital, Hund across the n pair
- BO = (bonding e⁻ – antibonding e⁻) / 2**
- Any unpaired e⁻ → **paramagnetic**; all paired → **diamagnetic**
- BO > 0 → stable, can form (H₂ 1 ✓; He₂ 0 x can't exist); **higher BO = stronger & shorter bond** (N₂ weaker/longer than N₂). σ2s + σ*2s cancel — you may count 2p only

Sp.	e ⁻	BO	Mag	Sp.	e ⁻	BO	Mag
N ₂	10	3	dia	O ₂	12	2	para (2↑)
N ₂ ⁺	9	2.5	para	O ₂ ⁺	11	2.5	para
N ₂ ⁻	11	2.5	para	O ₂ ⁻	13	1.5	para
F ₂ , Cl ₂	14	1	dia	Cl ₂ ⁺	13	1.5	para
B ₂	6	1	para	Cl ₂ ⁻	15	0.5	para

- Heteronuclear (NO, OF) → the O₂-type (right) ordering (more EN atom sits lower); period 3 (Cl₂, Na₂) → same shapes, n = 3

(26) WORD: Phase-Change Heat (gas ⇌ solid)



- Sketch the path; skip segments you don't cross (65 °C liq → 115 °C gas: heat liq 65→100, vaporize, heat gas 100→115)
- Sloped (one phase): **q = m(g)·Cs·ΔT** — flat (phase change, T constant): **q = n(mol)·ΔH**
- Sum; convert J → kJ. Cooling path (gas → solid): every q **negative** (ΔHcond = –ΔHvap, ΔHfreeze = –ΔHfus)
 - H₂O: C(ice) 2.09, C(liq) 4.184, C(steam) ≈ 2.0 J/g·°C; ΔHfus 6.02, ΔHvap 40.7 kJ/mol; MM 18.02
 - 36.0 g H₂O, 65 °C liq → 115 °C gas: 36×4.184×35 = 5.27 kJ; (36/18.02)×40.7 = 81.3 kJ; 36×2.01×15 = 1.09 kJ → **87.7 kJ**
 - Cooling: 125 g benzene (78.11) gas 425 K → liq 335 K (bp 353 K, mp 279 K – never freezes): 125×1.06×72 = 9.5 kJ; (125/78.11)×33.9 = 54.3 kJ; 125×1.73×18 = 3.9 kJ → **67.7 kJ removed**
 - Endo (+): melt, vaporize, sublime; Exo (–): freeze, condense, deposit; ΔHsub = ΔHfus + ΔHvap

(27) WORD: Raoult's Law (VP Lowering)

$$P_{\text{soln}} = \chi_{\text{solvent}} P_{\text{solvent}}^{\circ} \quad \Delta P = \chi_{\text{solute}} P_{\text{solvent}}^{\circ}$$
$$\chi_{\text{solvent}} = \frac{\text{mol solvent}}{\text{mol solvent} + i \cdot \text{mol solute}}$$

- mol solvent = g ÷ MM; mol solute = g ÷ MM (**x i if ionic**)
- χ(solvent) = mol solvent ÷ total mol — always < 1
- P(soln) = χ(solvent) × P^o; lowering ΔP = P^o – P(soln) = χ(solute)·P^o
 - 75.0 g citric acid (192.12) + 420 g H₂O, P^o 71.93 mmHg: 0.390 & 23.31 mol → χ = 0.9835 → **70.7 mmHg**
 - 200 g sucrose (342.3) + 350 g H₂O, P^o 17.5: 0.584 & 19.42 mol → χ = 0.9708 → **17.0 mmHg**

(28) WORD: Freezing ↓ / Boiling ↑ (van't Hoff)

$$\Delta T_f = i K_f m \quad \Delta T_b = i K_b m$$

- m = mol solute / **kg solvent**; H₂O: Kf 1.86, Kb 0.512 °C/m; Tf = T^of – ΔTf, Tb = T^ob + ΔTb
- i** = particles per formula unit: nonelectrolytes (glucose, sucrose, urea) 1; NaCl, KCl, NaOH, NH₄Cl 2; CaCl₂, (NH₄)₂SO₄ 3; Al(NO₃)₃, FeCl₃ 4 (measured i slightly less — ion pairing)
- Find i: **i = ΔTb / (Kb·m)** — 1.83 m boils 102.5 °C: 2.5/(0.512×1.83) = **2.7**
- Molar mass from fp: ΔTf → m = ΔTf/(i·Kf) → mol = m × kg → **MM = g/mol**: 20.0 g per 500 g benzene (T^of 5.444, Kf 5.12) freezes 3.77 °C: ΔT 1.674 → 0.327 m → 0.163 mol → **≈120 g/mol**
- Lowest fp, equal grams = most mol × i (small MM wins: NaOH > KCl, CaCl₂)

(29) WORD: Osmotic Pressure → Molar Mass

$$\pi = i M R T \Rightarrow M = \frac{\pi}{i R T}$$

- R = **0.08206** L·atm/mol·K; T in **K** (+273.15); n in **atm** (torr ÷ 760); i = 1 nonelectrolyte
- 1. M = n / (iRT)
- 2. mol = M × V(L)
- 3. **MM = g solute ÷ mol**
 - 15.2 mg in 150.0 mL, n 8.44 torr, 25 °C: M = (8.44/760)/(0.08206×298) = 4.54e–4 → 6.81e–5 mol → 0.0152/6.81e–5 = **223 g/mol**
 - n → fp: M from n → per 1 L: kg solvent = (1000·d – M·MM)/1000 → molality → ΔTf = i·Kf·m